

Machine Learning for Financial Models

Regression and Classification

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1 Introduction and Basic Concepts

In this section, we introduce the key concepts and terms used in these studies so that the data and results are easily interpreted by someone without a strong background in finance, like ourselves. We assume familiarity with supervised machine learning, including examples, features, targets, training sets, validation sets, and test sets. The models are implemented in Python with scikit-learn [3].

An **asset** is something with a value that can be bought or sold. A company share and a government bond are examples.

A **bond** is a loan that can be bought and sold. Buying a government bond means lending money to that government. Maturity is the date on which the loan is repaid. For a simple bond that pays 100 after one year and costs P today, its yield y satisfies

$$P = \frac{100}{1 + y}. \quad (1)$$

Here P is the current price and y is the yearly yield written as a decimal. If $P = 95$, then $y = 100/95 - 1 = 5.26\%$. The yield is calculated from the price and future payment, so a lower price gives a higher yield.

One **basis point** is 0.01 percentage point. A yield increase from 2.50% to 2.60% is 10 basis points. The **yield curve** is the set of yields for bonds with different maturities. We describe part of its shape using

$$s_t = y_{10,t} - y_{2,t}. \quad (2)$$

Here s_t is the yield-curve slope on date t , and $y_{10,t}$ and $y_{2,t}$ are the ten-year and two-year yields. If they are 3.0% and 2.5%, the slope is 0.5 percentage point, or 50 basis points.

A **loan default** means that a borrower has failed to make required payments under the lender's definition. The credit dataset supplies a binary target that records whether default occurred within twelve months after a loan was issued. A target value of 1 records default, and 0 records no default during that period.

All rows remain in chronological order. The earliest rows form the **training set**, which is used to fit model parameters. The following rows form the **validation set**, which is used to compare choices or check the model before final evaluation. The latest rows form the **test set**. This order matches the intended use of predicting later outcomes from earlier data.

The first project uses **quantile regression**. Ordinary regression predicts the average value associated with a set of features. Quantile regression predicts a selected percentile. For example, a 99th-percentile prediction is a threshold that should be exceeded on about 1% of later observations.

The second project uses **binary classification**. Logistic regression converts its fitted feature score into a probability between zero and one. Gradient boosting provides a nonlinear comparison in both projects by adding small decision trees in sequence. Each project gives the equation, training loss, and evaluation measures used in its analysis.

2 Project 1: German Government-Yield Forecasting

Aim. We use quantile regression to predict high thresholds for the next day's increase in the German ten-year government yield. We compare a historical percentile with a gradient-boosted regression model at the 97.5th and 99th percentiles.

2.1 Data and preprocessing

The input contains daily yields of current German federal securities published by the Deutsche Bundesbank [1]. The worksheet has 6,591 rows from 1 August 2000 to 14 July 2026. Each row is one published date. Its five original columns are described below.

date is the observation date. **yield_2y**, **yield_5y**, **yield_10y**, and **yield_30y** are the published yearly percentage yields for securities with approximately two, five, ten, and thirty years to maturity. Their ranges are -1.02% to 5.32% , -0.98% to 5.35% , -0.85% to 5.43% , and -0.50% to 6.27% . A negative yield can occur when a security's price is high enough that holding it until repayment gives a small loss.

We first calculate the daily change in the ten-year yield:

$$\Delta_t = 100(g_{10,t} - g_{10,t-1}). \quad (3)$$

Here Δ_t is the change on date t in basis points, and $g_{10,t}$ and $g_{10,t-1}$ are the published ten-year yields on consecutive dates. The target in row t is $y_t = \Delta_{t+1}$, the change on the next available date.

We calculate three yield-curve features:

$$a_t = 100(g_{10,t} - g_{2,t}), \quad b_t = 100(g_{30,t} - g_{10,t}), \quad k_t = 100(2g_{5,t} - g_{2,t} - g_{10,t}). \quad (4)$$

Here a_t is the ten-year minus two-year difference, b_t is the thirty-year minus ten-year difference, and k_t measures the position of the five-year yield relative to the two-year and ten-year yields. The factor 100 converts percentage-point differences into basis points.

Recent yield changes provide four more features:

$$v_{h,t} = \sqrt{\frac{1}{h} \sum_{j=0}^{h-1} \Delta_{t-j}^2}, \quad h \in \{1, 5, 20, 60\}. \quad (5)$$

Here $v_{h,t}$ is the root mean squared change over the latest h observations and j counts backwards from date t . The four windows represent the current day and approximately one week, one month, and three months. Squaring gives positive and negative changes the same sign. The square root returns the result to basis points.

The final ML dataset has eleven features. Four are $v_{1,t}$, $v_{5,t}$, $v_{20,t}$, and $v_{60,t}$. Four are the original two-year, five-year, ten-year, and thirty-year yields. Three are a_t , b_t , and k_t . These features describe recent movement size and the current shape of the yield curve. Every feature belongs to date t and the target belongs to the next date.

The historical comparison uses the selected percentile of the latest 1,000 ten-year yield changes. The first 1,000 changes therefore provide its initial history. The fitted ML model uses every earlier row available at the start of an evaluation year.

2.2 Analysis and results

The rows remain in date order. At the beginning of each year, we fit the model on all rows whose targets occur before that year. This gives 1,008 validation predictions from 2017 to 2020. After the model settings are fixed, the 1,408 predictions from 2021 onward form the test set.

For target y_i and predicted percentile q_i , quantile loss is

$$L_\tau(y_i, q_i) = \begin{cases} \tau(y_i - q_i), & y_i \geq q_i, \\ (1 - \tau)(q_i - y_i), & y_i < q_i. \end{cases} \quad (6)$$

Here τ is 0.975 or 0.99, y_i is the next-day yield change, and q_i is the predicted threshold. A 99th-percentile prediction that is one basis point below the observed change receives loss 0.99. A prediction one basis point above it receives loss 0.01. Training minimises the mean of this loss, which gives more weight to thresholds that are too low.

The gradient-boosting model starts with one constant prediction and adds 80 small decision trees. A tree divides feature values into ranges and assigns a correction within each final range. We use learning rate 0.04, at most 10 leaves per tree, at least 100 training rows per leaf, and an L_2 leaf penalty of 3. These limits keep individual trees small and avoid rules based on a few unusual observations.

An exception is a day on which the observed increase is above the predicted threshold. At the 99th percentile, 14.08 exceptions are expected among the 1,408 test forecasts. The historical method gives 20 exceptions, or 1.42%. Gradient boosting gives 23 exceptions, or 1.63%. Both thresholds are exceeded more often than the intended 1% rate.

Mean test quantile loss is 0.185 basis points for the historical method and 0.168 basis points for gradient boosting. The ML prediction has the smaller average quantile error, while the historical prediction has the exception rate closer to 1%. At the 97.5th percentile, the historical and ML methods give 55 and 46 exceptions. The expected number is 35.20. Their mean quantile losses are 0.366 and 0.325 basis points.

The validation results give another view of the comparison. At the 99th percentile, 10.08 exceptions are expected. The historical method gives 10 and gradient boosting gives 5. Their mean quantile losses are 0.131 and 0.106 basis points. At the 97.5th percentile, the two methods give 21 and 18 exceptions when 25.20 are expected.

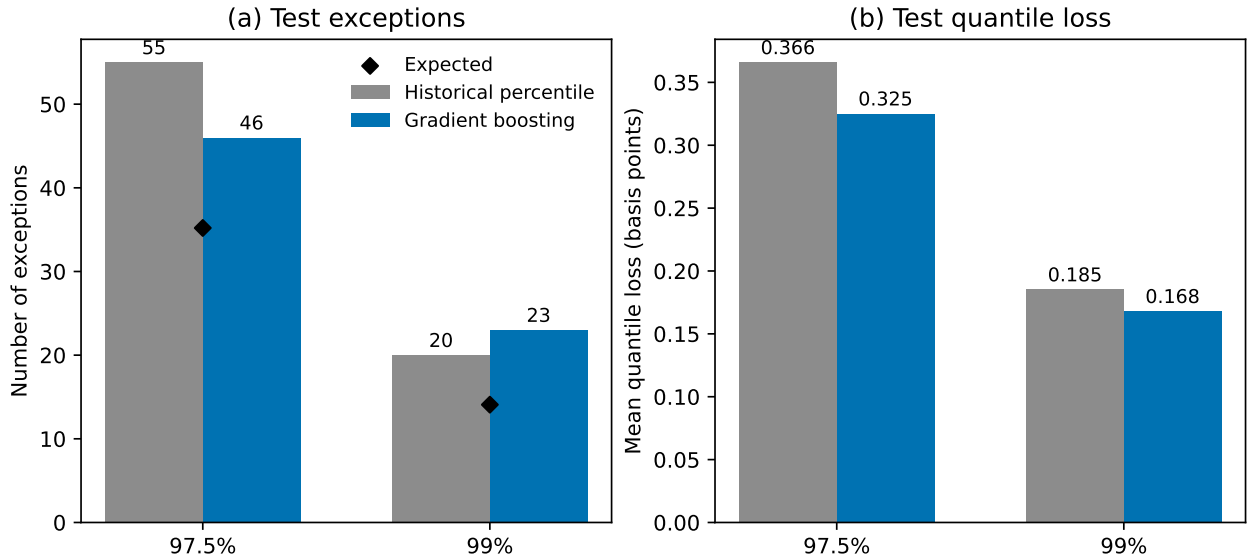


Figure 1: German ten-year yield results for the 97.5th- and 99th-percentile test forecasts. Panel (a) compares the observed exception counts from the historical percentile in grey and gradient-boosted quantile regression in blue. The black diamonds mark the expected counts of 35.20 and 14.08 among 1,408 forecasts. Panel (b) compares mean quantile loss, where a smaller value gives the smaller average quantile error.

The model is evaluated over several interest-rate conditions, including negative yields and the later

rise in rates. The test exception rates remain above the intended levels. Further work could repeat validation over several time windows, compare different history lengths, and calculate uncertainty around the exception rates. Any revised settings should be evaluated on later observations that were not used in the present comparison.

3 Project 2: Predicting Twelve-Month Loan Default

Aim. We use binary classification to predict whether a loan defaults within twelve months of issue. We compare features available from the loan request, features assigned during pricing, and the lender’s published risk rating.

3.1 Data and preprocessing

The source is the public Go & Grow loan workbook [2]. The worksheet contains 737,889 loan rows and 31 named columns. One row represents one loan. The columns describe the loan at issue, its later payments and status, and the published twelve-month outcome.

loan_id identifies a loan. **country** records the borrower’s country. **loan_issued_at** is the issue time. **early_repaid_at** records an early repayment time, and **is_early_repaid_within_14_days** records whether this happened within 14 days.

issued_amount is the original loan amount. **loan_status** is its recorded status.

loan_last_recorded_action_date_local

is the date of the latest recorded action. **initial_interest_rate** is the yearly rate assigned at issue. **nr_of_payments** is the planned number of payments.

principal_balance is unpaid principal. **principal_debt** is principal recorded as debt. **principal_paid_total**, **interest_paid_total**, **extra_interest_paid_total**, **late_fee_paid_total**, and **maintenance_fee_paid_total** are cumulative amounts received in the named categories.

is_default is the current default flag. **next_payment_nr** and **next_payment_date_local** describe the next scheduled payment. **debt_occured_date_local** gives the recorded debt date. **days_past_due_principal** is the number of days principal is late. **months_in_default** and **months_on_book** measure elapsed time in default and since issue. **loan_status_risk** is a status-based risk field, and **repaid_amount_total** is total repayment received.

initial_loan_duration is the original duration in months. **combined_income** is the recorded borrower income. **has_default_within_12_months** is the target supplied by the publisher: 1 means default within twelve months and 0 means no default within twelve months. **projected_npv_return** is the lender’s projected Net Present Value (NPV) return at issue. NPV expresses future payments in today’s value. **customer_risk_rating** is the lender’s grade from AA through High Risk (HR).

We predict at the issue date. Payment totals, present loan status, late days, debt dates, and similar later records are unknown at that time. We therefore select ten source columns: loan ID, country, issue date, issued amount, initial interest rate, initial duration, combined income, the twelve-month target, projected NPV return, and customer risk rating. Loan ID fixes row order and is not a model feature.

We retain loans issued from 1 October 2023 through 31 March 2025 in Estonia, Finland, Latvia, the Netherlands, or Denmark. Each retained row has a recorded value for the ten selected columns. We sort first by issue time and then by loan ID. This leaves 212,978 loans and 17,416 defaults, an overall rate of 8.18%.

We standardise issued amount, combined income, initial duration, initial interest rate, and projected NPV return using training-set statistics:

$$z_{i,j} = \frac{x_{i,j} - \mu_{j,\text{train}}}{s_{j,\text{train}}}. \quad (7)$$

Here $x_{i,j}$ is the original value of numerical feature j for loan i , $\mu_{j,\text{train}}$ is its training mean, and $s_{j,\text{train}}$ is its training sample standard deviation. The same means and standard deviations are

applied to validation, test, and Denmark loans. This puts the numerical features on comparable scales for logistic regression.

Country becomes one 0 or 1 column for each of the four countries present in training. The rating-only model also uses one 0 or 1 column per rating present in training. Each grade has its own fitted contribution, and loans with the same rating receive the same predicted probability in that model.

We use three feature sets. The **public feature set** contains issued amount, initial duration, combined income, and country. It becomes seven numerical columns after country encoding. The **public and pricing feature set** adds initial interest rate and projected NPV return, giving nine numerical columns. The **rating feature set** contains only the customer risk-rating indicators. All models predict the same twelve-month default target.

The main analysis uses 212,796 loans from Estonia, Finland, Latvia, and the Netherlands. The training set has 78,252 loans issued through 30 June 2024 and a default rate of 8.49%. The validation set has 48,657 loans issued from 1 July to 30 September 2024 and a default rate of 8.70%. The test set has 85,887 loans issued from 1 October 2024 through 31 March 2025 and a default rate of 7.53%. Denmark is absent from training. Its 182 test-period loans are kept as a separate check.

These are retrospective comparisons of issue-date cohorts. A historical deployment backtest would also require every training loan’s twelve-month outcome to be known before a later prediction was made.

3.2 Analysis and results

We fit three logistic-regression models, one for each feature set. For loan i , logistic regression predicts

$$\hat{p}_i = \frac{1}{1 + \exp[-(\beta_0 + x_i^\top \beta)]}. \quad (8)$$

Here $\hat{p}_i$ is the predicted default probability, β_0 is the intercept, x_i is the feature vector, and β contains the fitted coefficients. The coefficients are fitted by minimising binary log loss with an L_2 coefficient penalty:

$$-\frac{1}{N} \sum_{i=1}^N [Y_i \log(\hat{p}_i) + (1 - Y_i) \log(1 - \hat{p}_i)] + \lambda \sum_j \beta_j^2. \quad (9)$$

Here N is the number of training loans, Y_i is the observed 0 or 1 target, $\hat{p}_i$ is its predicted probability, and λ controls the coefficient penalty. Scikit-learn controls the penalty with C : smaller values give a stronger penalty. We use its default $C = 1$.

We also fit gradient boosting using the same standardised public and pricing inputs. It provides a nonlinear comparison using 250 small classification trees, learning rate 0.05, at most 15 leaves per tree, at least 100 training loans per leaf, and an L_2 leaf penalty of 1. We fix the random seed to 0 for repeatability. These settings are fixed for the model comparison.

ROC AUC measures ranking:

$$\text{ROC AUC} = P(s_D > s_N) + \frac{1}{2}P(s_D = s_N). \quad (10)$$

Here s_D is the score for a randomly selected defaulted loan, s_N is the score for a randomly selected nondefaulted loan, and P denotes probability. Tied scores receive half credit. A value of 0.5 represents random ordering and 1 represents perfect ordering.

The Brier score measures probability error:

$$\text{Brier} = \frac{1}{N} \sum_{i=1}^N (\hat{p}_i - Y_i)^2. \quad (11)$$

Here N is the number of evaluated loans, $\hat{p}_i$ is predicted probability, and Y_i is the observed target. A smaller value means a smaller mean squared probability error.

The public and pricing logistic model gives the highest test ROC AUC, 0.6909. The rating logistic model gives 0.6788, gradient boosting gives 0.6791, and public logistic regression gives 0.6158. Adding interest rate and projected return raises logistic-regression AUC by 0.0750 relative to the public model and by 0.0120 relative to the rating model.

The public and pricing logistic model also gives the lowest test Brier score, 0.06734. The corresponding values are 0.06771 for the rating logistic model, 0.06763 for gradient boosting, and 0.06893 for the public logistic model. Its mean predicted default probability is 8.35%, compared with an observed test default rate of 7.53%.

We divide the test loans into five approximately equal-sized groups ordered by the selected model's predicted probability. The lowest group has mean predicted probability 3.19% and observed default rate 2.06%. The middle group has predicted and observed rates of 6.88% and 6.98%. The highest group has mean predicted probability 17.24% and observed default rate 15.36%. The model overpredicts defaults overall, with closer agreement in the middle group.

Country checks use the observed-to-predicted ratio

$$O/P_g = \frac{\sum_{i \in g} Y_i}{\sum_{i \in g} \hat{p}_i}. \quad (12)$$

Here g is a country or time group, Y_i is the observed target, and $\hat{p}_i$ is predicted probability. A value of 1 means that predicted and observed default counts agree. Latvia has observed default rate 28.11%, mean prediction 23.70%, and ratio 1.19. Estonia has ratio 1.04. Finland and the Netherlands have ratios 0.83 and 0.86, respectively. The fourth quarter of 2024 has predicted and observed rates of 7.68% and 7.38%. The first quarter of 2025 has rates of 9.02% and 7.68%.

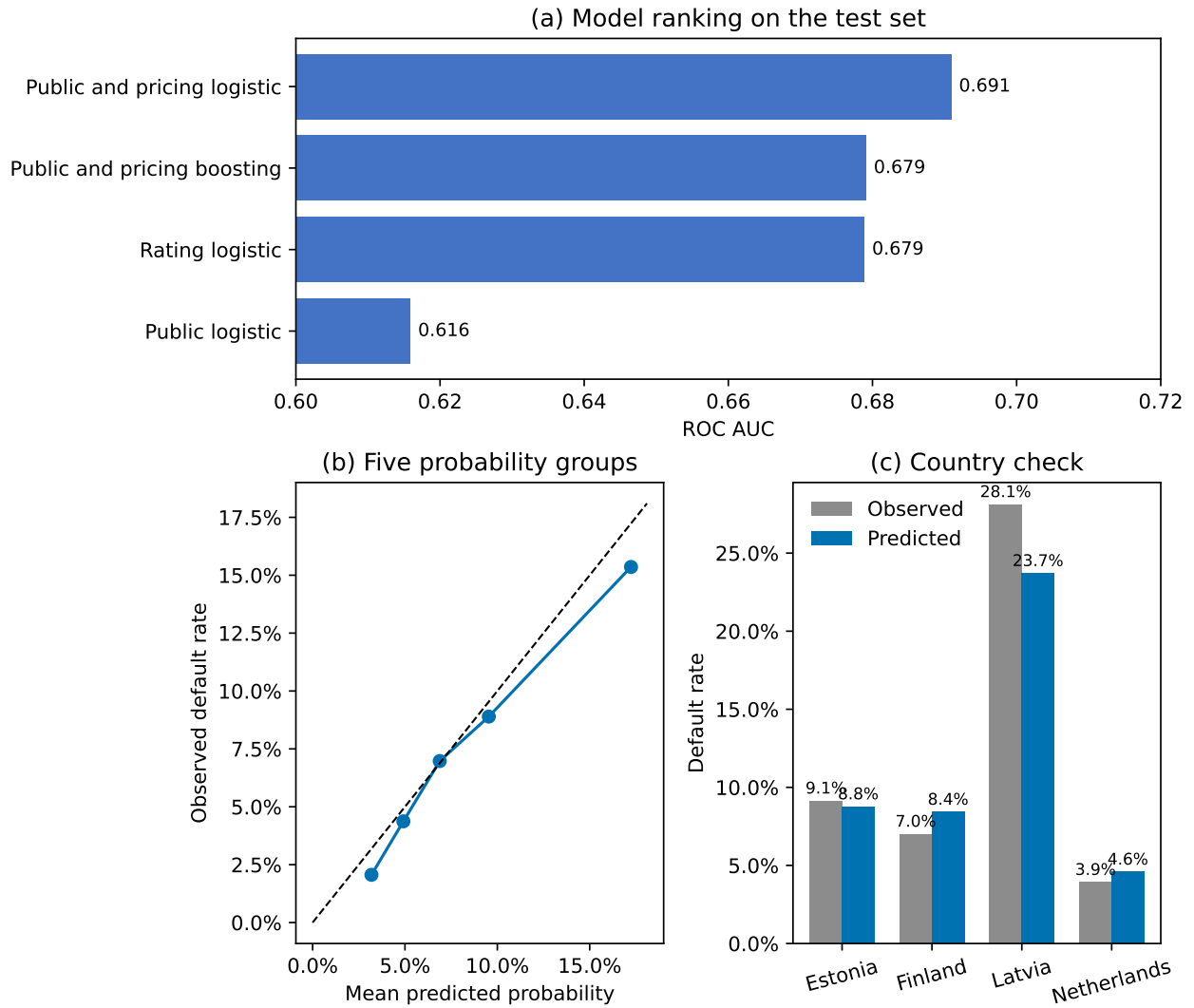


Figure 2: Credit results for the test period. Panel (a) compares ROC AUC for the four models. Public features are issued amount, duration, income, and country; pricing adds interest rate and projected return. Panel (b) checks the public and pricing logistic model across five approximately equal-sized probability groups. Each point compares a group’s mean predicted probability with its observed default rate. The dashed line marks equal predicted and observed rates. Panel (c) compares observed rates in grey with mean predicted rates in blue for the four countries present during training.

The separate Denmark group contains 182 loans, of which 72 default. Its observed rate is 39.56%, while the mean prediction is 9.76%. Denmark has no country coefficient because it does not occur in the training data. This result shows the risk in applying a model to a group absent from training.

The outcome covers twelve months after issue, and the analysis covers a short issue period. Later work could repeat the test on a longer period and examine more loans from countries absent during training. The present result shows binary classification, feature selection, chronological splitting, and probability evaluation on a real loan dataset.

References

- [1] Deutsche Bundesbank. Yields of current federal securities. Daily statistical series. Accessed 4 August 2026.
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